AP CALCULUS AB

Unit 1 • Lesson 1.13 notes

Lesson 1.13 • Removing Discontinuities

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Redefine a function at a point to remove a removable discontinuity.
- Find constants that make a piecewise function continuous.

Vocabulary. removable discontinuity, extended function, piecewise function, parameter.

1. Filling a hole

If $\lim_{x\to c} f(x) = L$ exists but $f(c) \neq L$ (or is undefined), defining $f(c) = L$ makes f continuous at c. Jump and infinite discontinuities cannot be removed this way.

Example 1

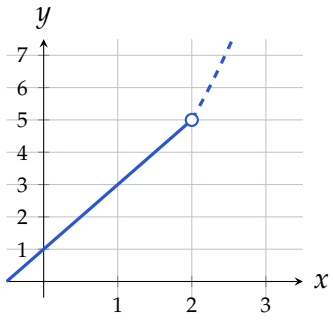
$f(x) = \dfrac{x^2 - x - 12}{x - 4}$, $x \neq 4$. What value of $f(4)$ makes f continuous?

Solution. $\dfrac{(x-4)(x+3)}{x-4} = x + 3 \to 7$, so define $f(4) = 7$.

2. Matching pieces

At a boundary point, set the left-hand limit, the right-hand limit and the function value equal.

Example 2 • One constant



$$f(x) = \begin{cases} 2x + 1, & x < 2 \\ x^2 + k, & x \geq 2. \end{cases}$$

Solution. Left: 5. Right and $f(2)$: $4 + k$. So $4 + k = 5$ and $k = 1$ (the dashed piece shows $x^2 + 1$).

Example 3 • Two constants

$$g(x) = \begin{cases} ax + b, & x < 1 \\ 5, & x = 1 \\ x^2 + bx, & x > 1 \end{cases}$$ is continuous at $x = 1$. Find a and b.

Solution. Right: $1 + b = 5$, so $b = 4$. Left: $a + b = 5$, so $a = 1$.

Common mistakes

- Setting only two of the three quantities equal when a separate value $f(c)$ is given.
- Trying to "remove" a jump discontinuity by redefining one point.

Quick check

1. Define $h(0)$ to make $h(x) = \frac{\sin 3x}{x}$ continuous at 0.
2. Find k so that $\begin{cases} kx - 1, & x < 3 \\ 2x + k, & x \geq 3 \end{cases}$ is continuous.

Answers. 1. $h(0) = 3$. 2. $3k - 1 = 6 + k$, so $k = \frac{7}{2}$.